ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure and Syllabus

(From Academic Session 2024-25 onwards)

B. TECH
INSTRUMENTATION ENGINEERING
4th SEMESTER

ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Course Structure
(From Academic Session 2024-25 onwards)

B. Tech 4th Semester: Instrumentation Engineering
Semester IV

Sl. No.	Course Code	Course Type	Course Title	Hours per Week			Credit	Marks	
				L	T	P	C	CE	ESE
Theory									
1	EI241401	PCC	Control Systems	3	0	0	3	30	70
2	IE241402	PCC	Industrial Instrumentation	3	0	0	3	30	70
3	EI241403	PCC	Microprocessors and Microcontrollers	3	0	0	3	30	70
4	IE241404	PCC	Modern Analytical Instrumentation	3	0	0	3	30	70
5	EI241405	PCC	Signals and Systems	3	0	0	3	30	70
6	HS241406	HSMC	Finance and Accounting	3	0	0	3	30	70
7	AU241407	AU	Environmental Science	3	0	0	0 (PP/NP)	100	-
Practical									
8	EI241411	PCC	Control Systems Lab	0	0	2	1	15	35
9	IE241412	PCC	Industrial Instrumentation Lab	0	0	2	1	15	35
10	EI241413	PCC	Microprocessors and Microcontrollers Lab	0	0	2	1	15	35
11	PR241415	PR	Micro Project (Skill Based)	0	0	4	2	15	35
TOTAL				**21**	**0**	**10**	**23**	**340**	**560**
Total Contact Hours per week: 31									
Total Credit: 23									

Course Code	Course Title	Hours per week L-T-P	Credit
EI241401	**Control Systems**	**3-0-0**	**3**

Prerequisites: Mathematics III

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: Distinguish between open loop and closed loop control systems
- **CO2**: Develop mathematical models of control systems and its components
- **CO3**: Analyze transient and steady-state behaviour of control systems
- **CO4**: Analyze stability of control systems using analytical and graphical techniques
- **CO5**: Design compensators for control systems

MODULE 1: Elementary concepts of control systems (4 Lectures)
Introduction to control systems, history and development of automatic control; open loop and closed loop control; linear and non-linear control, linear time invariant systems, continuous and discrete time analog and digital control systems; block diagram representation of control systems.

MODULE 2: Mathematical models of physical systems (7 Lectures)
Definition and properties of transfer function; open loop and closed loop transfer function, feedback characteristics of control systems, characteristic equation; transfer function of electrical and mechanical systems; analogous systems: force-voltage and force-current analogy; block diagram reduction technique and signal flow graph.

MODULE 3: Introduction to control system components (3 Lectures)
Controller components, potentiometer, AC/DC servomotors, tacho-generator, synchros, stepper motor.

MODULE 4: Time-domain analysis (7 Lectures)
Introduction to transient response; Test signals: impulse, step, ramp and parabolic signals; transient response of first and second-order control systems, transient response specifications; steady state errors, error constants, types of systems, generalized error constants.

MODULE 5: Stability analysis (5 Lectures)
Concept of control system stability; Routh-Hurwitz stability criterion; Root locus techniques, design of control systems using root locus techniques.

MODULE 6: Frequency response analysis (8 Lectures)
Frequency response and specifications; polar plot; stability analysis using Bode plot and Nyquist plot, Nyquist stability criterion.

MODULE 7: Introduction to controller design (6 Lectures)
The design problem, approaches to the design problem, preliminary considerations of classical design; selection and realization of basic compensators, lead, lag and lead-lag compensators, compensation techniques; PI, PD and PID controllers

Text Books/References:

1. Nagrath, I. J.& Gopal, M, Control Systems Engineering, New Age International Publishers, 8th Edition, 2025
2. Ogata, K., Modern Control Engineering, Pearson Education, 5th Edition, 2015
3. Kuo, Benjamin C, Automatic Control Systems, Prentice Hall of India, 7th Edition, 2013
4. Golnaraghi, F.&Kuo, Benjamin C., Automatic Control Systems, McGraw Hill Education (India) Pvt. Ltd., 10th Edition, 2018
5. Rao V. Dukkipati, Analysis and Design of Control systems using MATLAB, New Age International Publishers, 2nd Edition
6. NPTEL course as suggested by the department

Course Code	Course Title	Hours per week L-T-P	Credit
IE241402	**Industrial Instrumentation**	**3-0-0**	**3**

Prerequisites: Fundamentals of Instrumentation and Transducer.

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: Identify different methods of measuring pressure.
- **CO2**: Analyze the electrical and non-electrical methods of flow and level measurement.
- **CO3**: Analyze different methods of force, torque, velocity and acceleration measurement.
- **CO4**: Describe various moisture, humidity, viscosity, density measurement devices used in industrial environment.

MODULE 1: Pressure Measurement (8 Lectures)
Significance of pressure measurement in industry, Manometers, Diaphragm, Bellows, Bourdon tubes etc. Electrical Pressure measuring instruments, Vacuum measurement – Mcleod gauge, Pirani gauge, Knudsen gauge, Ionization gauge etc,Testing and calibration of pressure gauges: dead weight tester.

MODULE 2: Flow Measurement (6 Lectures)
Mechanical type - Head type, Area type, Mass flow meter, Electrical type – Electromagnetic, Ultrasonic, Hotwire Anemometers and Digital meters.

MODULE 3: Level Measurement (6 Lectures)
Significance of level measurement in industry, float type level indication, Level measurement using displacer and torque tube, Bubbler system. differential pressure method of level measurement, Electrical types of level gauges using resistance, capacitance, nuclear radiation and ultrasonic sensors.

MODULE 4: Force and Torque Measurement (6 Lectures)
Basic methods of force measurement, electromechanical load cells, mechanical load cells, piezoelectric force transducer, Strain gauge torque meter, torsion bar torque meter, Inductive torque meter.

MODULE 5: Displacement, Velocity acceleration and Vibration measurement (8 Lectures)
Displacement, Velocity measurement (linear and angular), rotational speed measurement- drag cup tachometer, worm gear type, stroboscope, accelerometer-working principle, types, Relative and absolute (Seismic) accelerometer, Piezo-electric accelerometer, Bonded strain gauge type accelerometer.

MODULE 6: Moisture, humidity, viscosity, density measurement (6 Lectures)
Hydrometer - continuous weight measurement, liquid densitometer - float principle, air pressure balanced method, using gamma rays - gas density measurements – gas specific gravity measurements –Viscosity: Saybolt viscometer – Rotameter type and Torque type viscometers – Consistency Meters Humidity: Dry and wet bulb psychrometers – Resistive and capacitive type hygrometers – Dew cell. Moisture: Different methods of moisture measurements -Thermal, Conductivity and Capacitive sensors, Microwave, IR and NMR sensors, Moisture measurement in solids.

Text Books/References:

1. Ernest Doeblin, Measurement Systems: Application and Design, McGraw Hill. 5th Edition 2004
2. D Patranabis, Principles of Industrial Instrumentation, McGraw Hill., 3rd Edition 2017
3. B.EJones, Instrument Technology (vol-1&Vol2), Butterworth-Heinemann, 4th Edition 2016
4. B C Nakra and K K Chaudhry, Instrumentation Measurement and Analysis, McGraw Hill., 4t Edition 2016
5. K. Krishnaswamy, Industrial Instrumentation, NEW AGE, 2nd Edition 2020

Course Code	Course Title	Hours per week L-T-P	Credit
EI241403	**Microprocessors and Microcontrollers**	**3-0-0**	**3**

Prerequisites: Analog electronics and Digital Electronics

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: Explain the basic architecture and operation of 8-bit 8085 microprocessors
- **CO2:** Develop programs in 8085 assembly language for academic and industrial applications
- **CO3:** Interfacing peripheral devices to the 8085 microprocessors
- **CO4:** Describe basic architecture of 8086 16-bit microprocessor
- **CO5:** Design simple microcontroller-based applications using compatible peripherals

MODULE 1: Architecture of Microprocessors (8 Lectures)
Introduction to microprocessors, microcomputer and microcontroller, microprocessors architecture and its operations, memory classification and interfacing, 8085 microprocessor architecture, pins and signals, memory mapping and address decoding, Interfacing I/O devices with peripheral I/O and memory mapped I/O.

MODULE 2: Programming of 8085 (10 Lectures)
Data Transfer operations, Arithmetic operations, Logic Operations, Branch operation, 8085 instruction format and, addressing modes; assembly language programming of 8085. Programming techniques: looping, counting and indexing, delay routines. Instruction cycle, machine cycle, T-state, bus timing diagrams. Stack and subroutine usage, 8085 interrupt process, software and hardware interrupts, their priorities, concepts of Direct Memory Access.

MODULE 3: Interfacing Devices (6 Lectures)
8255, 8253 and 8279 Programmable Interfacing devices, various modes of operation and interfacing to 8085, designing simple systems for interfacing 8085 to I/O devices using PPIs

MODULE 4: 8086 Architecture (6 Lectures)
Architecture, pin description, physical address, segmentation, memory organization, addressing modes, basic programming.

MODULE 5: Introduction to Microcontrollers (10 Lectures)
Overview of 8051 Microcontroller, Architecture, I/O Ports, PSW and Flag Bits, 8051 register banks and stack, internal memory organization of 8051, types of special function registers and their uses in 8051, addressing modes and instruction set of 8051, simple programs. arithmetic, logic instructions and programs, jump, loop and call instructions.

Text Books/References:

1. Ramesh Gaonkar, Microprocessor Architecture, Programming and Applications with the 8085, Penram International PublishingPvt. Ltd, 2013
2. Muhammad Ali Mazidi, Janice Gillispie Mazidi and Rolin D. McKinley, The 8051 Microcontroller and Embedded Systems using Assembly and C, PearsonEducation 2nd Edition, 2008
3. Doughlas.V.Hall, Microprocessor and Interfacing: Programming and Hardware, McGraw Hill, 1991
4. Manish K Patel, The 8051 microcontroller Based Embedded Systems, McGraw Hill, 2014
5. Sunil Mathur, Microprocessor 8086: Architecture, Programming and Interfacing, PHI, 2010
6. NPTEL course as suggested by the department

Course Code	Course Title	Hours per week L-T-P	Credit
IE241404	**Modern Analytical Instrumentation**	**3-0-0**	**3**

Prerequisites: Chemistry, Fundamentals of Instrumentation and Transducers.

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: Analyze different techniques of spectrometry to find the composition of substance, their molecular structure etc.
- **CO2:** Explain various sampling techniques of liquid and gases; compare different gas analyzer and chromatographs.
- **CO3**: Estimate the content of hydrocarbons, carbon mono oxide and other harmful gases to check the extent of air pollution.
- **CO4**: Explain various X-ray spectroscopy and NMR spectroscopy techniques
- **CO5**: Explain various pH Meters and Dissolved Component Analyzers.

MODULE 1: Colorimetry and Spectrophotometry (8 Lectures)
Beer-Lambert law – Colorimeters, application of Colorimeters – Molecular UV-Vis absorption Spectroscopy, Application in qualitative and quantitative analysis, Molecular fluorescence, phosphorescence and chemiluminescence spectroscopy. UV-Vis Sources and detectors, IR absorption spectroscopy qualitative and quantitative analysis, – Atomic spectroscopy, atomic absorption types, atomic fluorescence types – Various types of the spectroscopy and their applications.

MODULE 2: Chromatography (8 Lectures)
Gas chromatography- Gas liquid and Gas solid type, High Pressure Liquid chromatography (HPLC), Partition Chromatography, Absorption and partition chromatography, Ion-exchange chromatography, Size exclusion chromatography, Applications of chromatography.

MODULE 3: pH Meters and Dissolved Component Analyzers (8 Lectures)
Principle of pH measurement – Glass electrodes – Hydrogen electrodes – Reference electrodes, selective ion electrodes and ammonia electrodes – Biosensors – Dissolved oxygen analyzer –Sodium analyzer – Silicon analyzer.

MODULE 4: Industrial Gas Analyzers and Pollution Monitoring Instruments (8 Lectures)
Types of gas analyzers – Oxygen, NO2 and H2S types, IR analyzers – Thermal conductivity analyzers – Analysis based on ionization of gases – Air pollution due to carbon monoxide, hydrocarbons – Nitrogen oxides – Sulphur-dioxide estimation – Dust and smoke measurements

MODULE 5: X-ray spectroscopy and NMR spectroscopy (8 Lectures)
X-ray spectroscopy, application, NMR - Basic principle, concept of chemical shift, NMR spectrometer, Applications, Mass spectrometers, Applications, GM counter, Proportional counter.

Text Books/References:

1. R.S. Khandpur, Handbook of Analytical Instruments, McGraw Hill., 3rd Edition 2015
2. H.H. Willard, L.L. Merritt, J.A. Dean, and F.A. Settle, Instrumental Methods of Analysis, CBS, 7th Edition 2004
3. R .D. Braun, Introduction to Instrumental Analysis, Pharma MED, 2012
4. G.W. Ewing, Instrumental Methods of Analysis, McGraw Hill., 3rd Edition 1969
5. D.A. Skoog, and D.M. West, Principles of Instrumental Analysis, Saunders, 2nd Edition 2020

Course Code	Course Title	Hours per week L-T-P	Credit
EI241405	**Signals and Systems**	**3-0-0-3**	**3**

Prerequisites: Mathematics for Engineering Systems

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: Identify various types of signals in continuous-time and discrete-time domain
- **CO2**: Illustrate Linear Time Invariant (LTI) systems and its properties
- **CO3**: Apply Fourier Series, Fourier Transform and z-transform techniques in signal processing of LTI systems
- **CO4**: Apply sampling techniques in signal processing
- **CO5**: Design digital filters using z-transform technique

MODULE 1: Introduction to signals and systems (5 Lectures)

Continuous-time and discrete-time signals, signal energy & power, even & odd signals, periodic signals, exponential and sinusoidal signals; transformation of independent variables: time-shift, time-reversal, time-scaling; unit step and unit impulse functions, Discrete-Time (DT) and Continuous-Time (CT) systems.

MODULE 2: Linear-Time-Invariant (LTI) Systems (8 Lectures)

DT & CT LTI systems: convolution sum, convolution integral; properties of LTI systems; LTI system with and without memory; Invertibility, causality and stability of LTI systems; LTI systems described by differential and difference equations and block diagrams.

MODULE 3: Fourier Series Analysis of Signals (7 Lectures)

Response of LTI system to CT and DT complex exponentials, Fourier series representation of CT and DT periodic signals, properties of CT and DT Fourier series, convergence of Fourier series.

MODULE 4: Fourier Transform Analysis of Signals (8 Lectures)

Fourier transform of DT and CT aperiodic and periodic signals, properties of CT and DT Fourier Transforms, convergence of Fourier Transforms, Inverse Fourier Transforms; LTI system characterized by linear constant-coefficient difference and differential equations.

MODULE 5: Sampling (4 Lectures)

Representation of a CT signal by its samples, Sampling Theorem; Concept of under-sampling, reconstruction of signals from its samples.

MODULE 6: Z-Transform (8 Lectures)

Definition, Region of Convergence (ROC), z-transform and inverse z-transform of signals, properties of z-transform, analysis and characterization of LTI systems using z-transforms, system functions and application to system analysis, FIR and IIR systems, realization of FIR and IIR filters.

Text Books/References:

1. Oppenheim, A. V., Willsky, A. S., Nawab, S. H., Signals and Systems, Prentice Hall India, 2nd Edition, 2006
2. Rawat, T. K., Signals and Systems, Oxford University Press, 1st Edition, 2010
3. Proakis, J. G. & Manolakis, D. G., Digital Signal Processing-Principles, algorithms and applications, Prentice Hall India, 3rd Edition, 2003
4. Nagoor Kani, A., Signals and Systems, McGraw Hill Education (India), 1st Edition, 2010
5. Robert, M. J., Signals and Systems, Tata McGraw Hill, 3rd Edition, 2004
6. Mitra, S. K., Digital Signal Processing-a computer-based approach, Tata McGraw Hill, 3rd Edition, 2008.

Course Code	Course Title	Hours per week L-T-P	Credit
HS241406	**Finance and Accounting**	**3-0-0**	**3**

Prerequisites: High School Education
Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Develop a strong foundation in accounting principles, including the classification of accounts, the Double Entry System, the application of the Golden Rules of Debit and Credit, record financial transactions in journals, post them into ledgers, and prepare a Trial Balance.
- **CO2:** Understand the role of subsidiary books in accounting, including different types of cash
- books and their preparation; learn to reconcile differences between the Cash Book and Pass Book balances and prepare a Bank Reconciliation Statement
- **CO3:** Differentiate between capital and revenue expenditure, understand the concepts of bad debts and provisions for doubtful debts and gain the skills to prepare final accounts by incorporating necessary adjustments.
- **CO4:** Explore the concepts of savings and investment, including their benefits and differences, identify various investment avenues in physical and financial assets and understand the distinction between trading and investment to make informed financial decisions.
- **CO5:** Have a comprehensive understanding of financial markets, covering money and capital markets, key financial instruments, IPOs, stock exchanges, and mutual funds, and analyze the structure and functions of financial markets and assess different financial instruments used for investment.
- **CO6:** Acquire knowledge and skills in stock market investment and analysis, applying fundamental and technical analysis to evaluate stocks, learn to interpret stock market charts and patterns, assess stock price movements, and utilize various analytical tools for informed investment decisions.

MODULE 1: Introduction to Accounting
Concept and classification of Accounts, Transaction, Double Entry System of Book Keeping, Golden rules of Debit and Credit, Journal- Definition, Advantages, Procedure of Journalising, Ledger, Advantages, Rules regarding Posting, Balancing of ledger accounts, Trial Balance- Definition, Objectives, Procedure of preparation.

MODULE 2: Subsidiary Books, Cash Book and Bank Reconciliation
Name of Subsidiary Books, Cash Book- Definition, Advantages, Objectives, Types of Cash Book, Preparation of different types of Cash Books, Bank
Reconciliation Statement, Reasons of disagreement between Cash Book with Pass Book balance, Preparation of bank Reconciliation Statement.

MODULE 3: Expenditures, Bad Debts and Final Accounts
Concept of Capital Expenditure and Revenue Expenditure, Bad Debts,
Provision for Bad and Doubtful debts, Final Accounts: Preparation of Trading Account, Profit and Loss account and Balance Sheet with adjustments.

MODULE 4: Introduction to Saving and Investment
Savings: Concept, Benefits and saving alternatives, Disadvantage of Saving – Time Value of Money. Investment: Meaning, Advantage, Investment Avenues – Investment in Physical and financial assets, Difference between Saving and Investment; Trading- Meaning, Difference between Trading and Investment.

MODULE 5: Financial Market and Instruments.

Financial Markets and Instruments: Money Market, Capital Market, Money Market Instruments and features. Capital Markets and Features, New Issue Market and Stock Market; New Issue market-Initial Public Offering , Market Players, Secondary Market :Equity and Debentures, BSE and NSE, Index and its uses; Mutual Funds: Meaning and Types.

MODULE 6: Stock Market Investment and Analysis

Stock Market Investment: Fundamental Analysis- Economic Analysis-Industry Analysis and Technical Analysis - Tools of technical analysis, Understanding various types of charts and patterns-(Line chart, Bar Chart, Candlestick Chart Point and Figure Chart)

Textbooks/Reference Books

1. Dam, B. B.Sarda, R.A. Barman, R. Kalita, B., Theory and Practice of Accountancy- Vol. I, Gayatri Publication, 2019
2. Goel, D.K Goel, R. Goel,S. Accountancy, Avichal publishing company, 2020
3. Kevin, S., Security Analysis and Portfolio Management, Prentice Hall India., 2022
4. Pathak,B., Indian Financial System, Pearson Education, 2024
5. Chandra, P., The Invesment Game: How To win, Tata McGrawHill, 2012

Course Code	Course Title	Hours per week L-T-P	Credit
AU241407	**Environmental Science**	**3-0-0**	**0**

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Relate environment and its associated problems
- **CO2:** Explain basic ethics of living and protection of other organisms
- **CO3:** Applying sustainable development principles

MODULE 1: Introduction
Environment and Ecology, Objectives of ecological study, Aspects of Ecology, Autecology, Synecology; Ecosystem, Structural and functional attributes of an ecosystem, Food chain and foodweb, Energy flow, Bio geochemical cycles

MODULE 2: Land: Use and Abuse
Land use: Impact of land–use on environmental quality, Land degradation, Control of land degradation, Wasteland, Wet lands

MODULE 3: Water Pollution
Introduction: Water quality standards, Water pollution

MODULE 4: Air Pollution
Introduction, Air pollution system, Air pollutants, Air pollution laws

MODULE 5: Noise Pollution
Sources of noise pollution, Effects of noise, Physical effects, Physiological effects

MODULE 6: Control of Pollutions
Control of water pollution and management, Water pollution legislations, Water quality management in surface water; Control of air pollution, source correction method, Pollution control equipment; controls of Noise pollution

Textbooks/Reference Books

1. Dr Suresh Dhameja, Environmental engineering and management, 2010
2. Dr B.S. Chauhan, Environmental studies
3. Henry and Hence, Environmental science and engineering
4. Dr Susmitha Baskar, Environmental studies for undergraduate course
5. Clair Sawyer, Chemistry for environmental engineering and science

Course Code	Course Title	Hours per week L-T-P	Credit
EI241411	**Control Systems Lab**	**0-0-2**	**1**

Prerequisites: Linear Control Systems

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: Solve control system problems using MATLAB software
- **CO2**: Analyze the response of control system by measuring relevant parameters
- **CO3**: Interpret the role of various components in control system
- **CO4**: Compare theoretical predictions with experimental results to resolve any apparent differences

Module No.	Description	Contact hours
PART I:	Use of MATLAB software for solving Control System Problems	10
PART II:	To perform experiments on control systems using experimental kits **LIST OF EXPERIMENTS:** 1. Light Intensity Control Systems 2. DC Position Control Systems 3. Potentiometer Error Detector 4. Speed-Torque Characteristics of AC Servomotor 5. Synchro-Transmitter Control Transformer pair as an Error Detector	10
	Total Contact Hours	20

Textbooks/Reference Books

1. D Hanselman, B. Littlefield, Mastering MATLAB, Pearson Education, 1st Edition.
2. Rao V. Dukkipati, Analysis and Design of control systems using MATLAB, New Age International Publishers, 2nd Edition.

Course Code	Course Title	Hours per week L-T-P	Credit
IE241412	**Industrial Instrumentation Lab**	**0-0-2**	**1**

Prerequisites: Industrial Instrumentation

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: Study Various applications of different types of transducers.
- **CO2**: Compare the efficiency of different types of transducers in the measurement of same physical parameter.
- **CO3**: Develop laboratory skill in handling various associated measuring instruments

List of Experiments:

Module No.	Description	Contact hours
1	To study Pressure Measurement by Piezo-Resistive Sensor.	2
2	To study Measurement of Liquid Level by Capacitive Transducer	2
3	To study Fore Measurement by Load Cell	2
4	To study Measurement of Speed by Proximity Sensor.	2
5	To study flow Measurement using Hot Wire Anemometer.	2
6	To study Measurement of Light Intensity by Photo Diode.	2
7	Measurement of Speed by Electromagnetic Pickup & Photo Electric Pickup	4
8	To study Torque Measurement by strain gauge.	2
9	Study of Stroboscope	2
	Total Contact Hours	20

Textbooks/Reference Books

1. Ernest O Doeblin, Measurement Systems: Application and Design, McGraw Hill., 5th Edition 2004
2. D Patranabis, Principles of Industrial Instrumentation, McGraw Hill., 3rd Edition 2017
3. B.E Jones, Instrument Technology (vol-1&Vol2), Butterworth-Heinemann, 4th Edition 2016
4. B C Nakra and K K Chaudhry, Instrumentation Measurement and Analysis, McGraw Hill., 4th Edition 2016
5. K. Krishnaswamy, Industrial Instrumentation, NEW AGE, 2nd Edition 2020

Course Code	Course Title	Hours per week L-T-P	Credit
EI241413	**Microprocessors and Microcontrollers Lab**	**0-0-2**	**1**

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: Understand the architecture and the instruction set of 8085 microprocessors
- **CO2**: Design and develop programs using 8085 microprocessor kit and 8051 microcontrollers
- **CO3**: Write an engineering report in an organized way

Experiment	Description	Contact hours
1	**Familiarization of the Microprocessor Kit:** (a) editing a program (b) Verifying the program (c) Executing the program and verifying the outcome of the program.	2
2	**Programming of 8085:** Developing and testing simple program for data transfer, block transfer, sorting of data, arithmetic operations, interrupt,	6
3	Interfacing 8255, traffic light control system	4
4	**Programming of 8051 using embedded C:** Develop simple programs for implementation in 8051 systems	8
	Total Contact Hours	20

Textbooks/Reference Books

1. Ramesh Gaonkar, Microprocessor Architecture, Programming and Applications with the 8085, Penram International PublishingPvt. Ltd, 2013

Course Code	Course Title	Hours per week L-T-P	Credit
PR241415	**Micro Project (Skill Based)**	**0-0-4**	**2**

Prerequisites: Knowledge of basic science

Objectives:

- To identify problems based on societal or research needs
- To acquaint with the process of solving the problem in a group
- To apply engineering knowledge to attempt solutions to the problems

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: Analyze engineering problems and their solutions
- **CO2**: Apply modern engineering tools to attempt solutions to the problems
- **CO3**: Adapt with computational thinking and automation skills
- **CO4**: Realize the need of professional communications, team work and self-learning

General Guidelines:

1. Students shall form a group of 3 to 4 students.
2. Each group shall conduct field survey or literature survey in consultation with faculty supervisor/mentor or internal project committee of faculty members to identify the needs of the society.
3. Each group shall formulate the problems on the basis of their survey and submit the implementation plan to the concerned faculty supervisor/mentor or the committee.
4. A log book is to be maintained by each group, wherein the group can record the work progress. The supervisor shall monitor the progress.
5. The faculty supervisor/mentor or the committee may give inputs to the students during the project activities, however, focus shall be on self-learning.
6. Students in a group shall understand problem effectively, propose multiple solutions and select the best solution in consultation with faculty supervisor/mentor or the committee.
7. Students shall convert the best solution into a working model/prototype using various components of their domain areas.
8. Students shall demonstrate and validate the model with proper justification and a report is to be compiled in a standard format/template

Assessment Criteria:

The Micro Project shall be assessed based on the following criteria:

Sl No.	Component	Weightage
1	Proper identification of the problem	20
2	Quality of survey	15
3	Clarity of problem definition	10
4	Presence of innovation	5
5	Effective use of modern engineering tools	10
6	Cost effectiveness	5
7	Societal impact	10
8	Functioning of the working model	15
9	Fulfilment of standard engineering norms	5
10	Clarity in written and oral communication	5
		100

Examinations:

Sl No.	Type of examination/evaluation	Marks (50)
1	Mid-term presentation on Progress	10
2	Demonstration of the project	10
3	End-term presentation	25
4	Viva-voce examination	5

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